Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

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Ringkasan

PENDAHULUAN
TINJAUAN PUSTAKA
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Your Name Your Name
200XXYYZZ **Assignment 1** CS 310-002
Fall 2019 1

Example Problem 1

Answer to the problem goes here. Use a line per sentence. Leave a blank space to start a new paragraph. Next, an example typesetting mathematics in LaTeX.

Showing that equation $a + b = \frac{c}{d}$ in evidence: a + b = c $\frac{}{d}$

Note that equation was automatically numbered. If you prefer not numbered equations, see the next example.

Example Problem 2

Showing that $\neg(p \to q)$ and $p \wedge \neg q$ are logically equivalent. $\neg(p \to q) \equiv \neg(\neg p \vee q)$
$\equiv \neg(\neg p \vee q)$
$\equiv \neg(\neg p) \wedge \neg(q)$
$\equiv p \wedge \neg q$

Note that & is where the equations align.

Example Problem 3

Constructing the *Truth Table* of $(p \to q) \wedge (\neg p \leftrightarrow q)$ in Table 1:

Tabel 1: Caption here. Leave it blank if you will not refer it.

p	q	$\neg p$	$p \to q$	$\neg p \leftrightarrow q$	$(p \to q) \wedge (\neg p \leftrightarrow q)$
T	T	F	T	F	F
T	F	F	F	T	F
F	T	T	T	T	T
F	F	T	T	F	F

Example Problem 4

[a)]"There is a student in Gryffindor who has taken all elective classes." Solution: $\exists x \forall y \forall z(H(x, Gryffindor) \wedge P(x, y))$ where

1. $H(x, z)$ is "x is of z house"
   $P(x, y)$ is "x has taken y,"
   the domain for x consists of all students in Hogwarts
   the domain for y consists of all elective classes,
   and the domain for z consists of all Hogwarts houses.

2. Give a direct proof of the theorem "If n is an odd integer, then n^2 is odd."
   Solution:
   [1.] $\forall n(P(n) \to Q(n))$, where
   (a) $P(n)$ is "n is an odd integer" and
   $Q(n)$ is "n^2 is odd."
   (b) Assume $P(n)$ is true.
   (c) By definition, an odd integer is $n = 2k + 1$, where k is some integer.
   (d) $\mathrm{n}^2 = (2k + 1)^2$
   $= 4k^2 + 4k + 1$
   $= 2(2k^2 + 2k) + 1$
   (e) n^2 is an odd integer.

3. Let $A = \{1, 2, 3\}$ and $B = \{1, 2, 3, \{1, 2, 3\}\}$:
   Then, $A \in B$ and $A \subseteq B$.

4. Let $A = \{1, 3, 5\}$, $B = \{1, 2, 3, \}$, and universe $U = \{1, 2, 3, 4, 5\}$: A $\cup B = \{1, 2, 3, 5\}$,
   $A \cap B = \{1, 3\}$,
   $A - B = \{5\}$,
   $\bar{A} = \{2, 4\}$,
   $A - A = \emptyset$.